FOP – Lab – Task 1.1

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Problem:

Create a simple tip calculator. The program should prompt for a bill amount and a tip rate. The program must compute the tip and then display both the tip and the total amount of the bill.

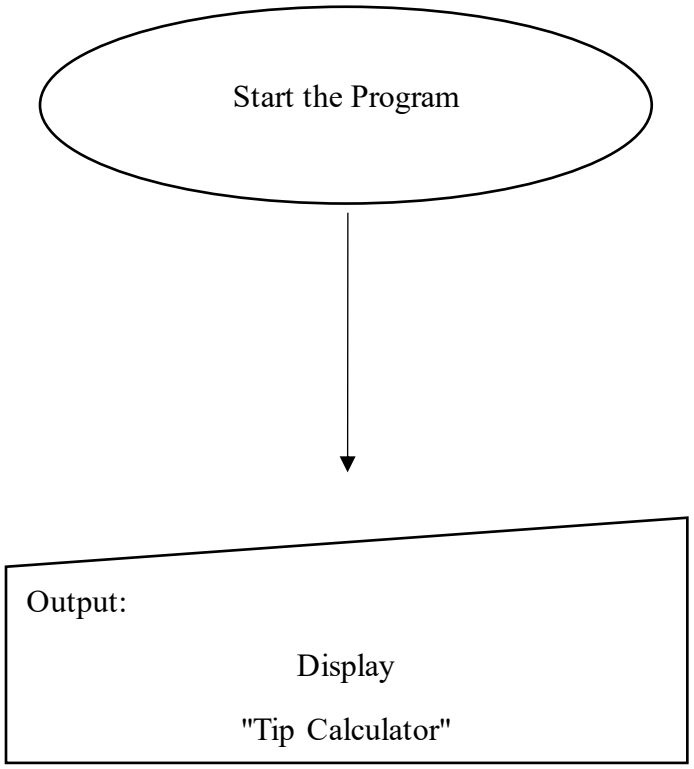
When you've finished writing the basic version of the program, try tackling some additional challenges:

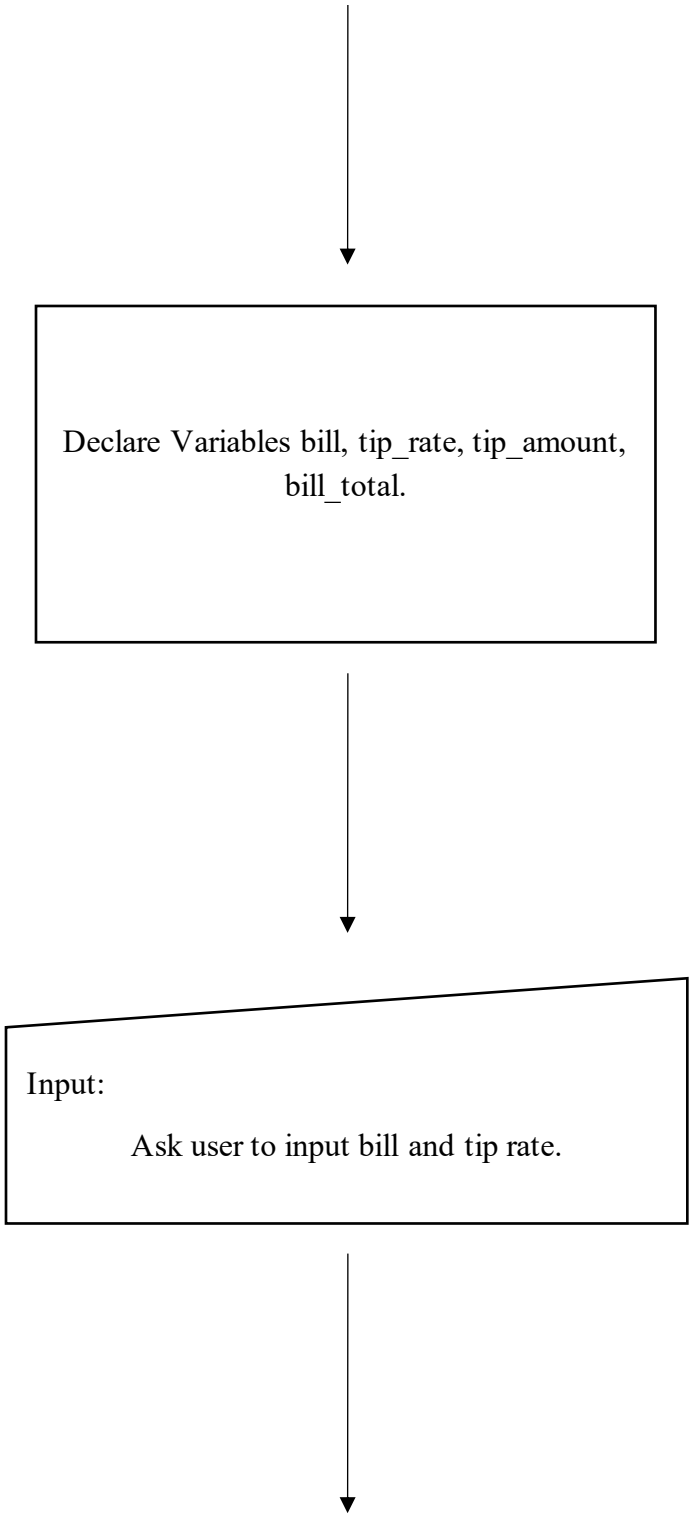
Ensure that the user can enter only numbers for the bill amount and the tip rate. If the user enters non-numeric values, display an appropriate message, and exit the program.
Here's a test plan as an example
Expected result: Please enter a valid number for the bill amount
Note: Don't allow the user to enter a negative number.
Convert the above requirement into Specifications (Input. Output, processing)

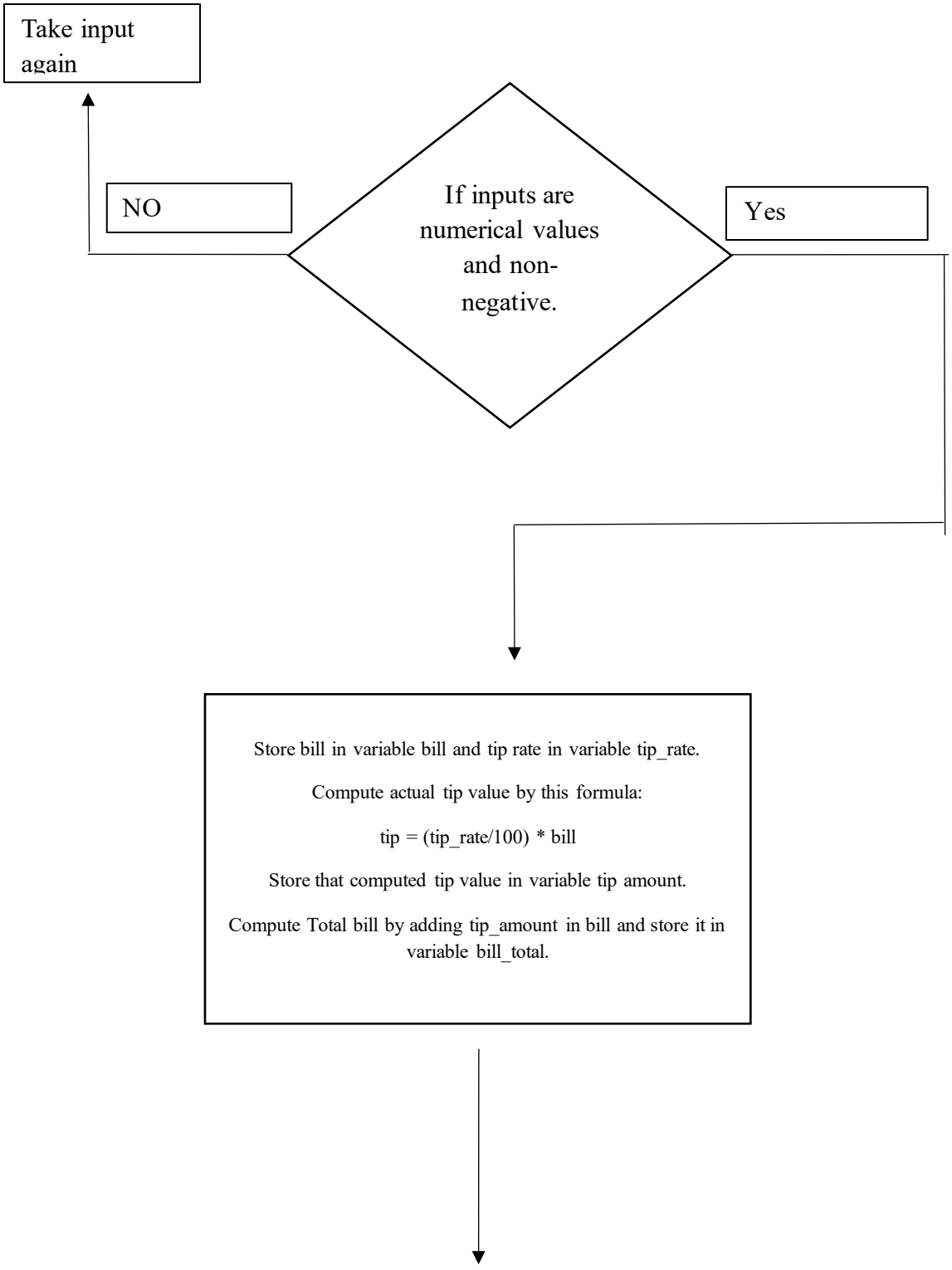
Algorithm:
Step 1:
Program Starts and Displays "Tip Calculator".
Step 2:
Declare variables bill, tip_rate, tip_amount, bill_total.
Step 3:
Ask user to input bill and tip rate.

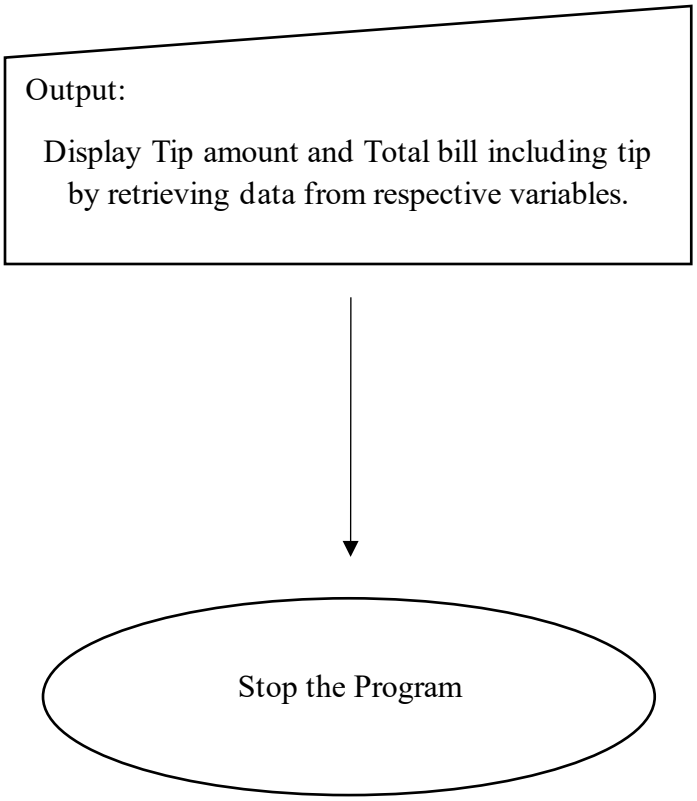
Step 4:
If the user enters a negative integer or any non-integer value. Then display “Invalid Input” and goto step 3 and ask the user to input again.
Else proceed to step 5.
Step 5:
Store bill in variable bill and tip rate in variable tip_rate.
Step 6:
Compute actual tip value by this formula:
tip = (tip_rate/100) * bill
Store that computed tip value in variable tip amount.
Step 7:
Compute Total bill by adding tip_amount in bill and store it in variable bill_total.
Step 8:
Display Tip amount and Total bill including tip by retrieving data from respective variables.
Step 9:
Stop the program.

Flowchart:




Declare Variables bill, tip_rate, tip_amount, bill_total.
Input:
Ask user to input bill and tip rate.


Take input again
NO
If inputs are numerical values and non-negative.
Yes
Store bill in variable bill and tip rate in variable tip_rate.
Compute actual tip value by this formula:
tip = (tip_rate/100) * bill
Store that computed tip value in variable tip amount.
Compute Total bill by adding tip_amount in bill and store it in variable bill_total.


Output:
Display Tip amount and Total bill including tip by retrieving data from respective variables.
Stop the Program